UART RTL-to-GDSII
RESULTS SUMMARY

Extracted from the supplied SCL summer internship report | Results only — no personal or administrative report pages

1. RTL functional simulation

Source: internship report, page 24, Figure 5.2 (GTKWave pre-synthesis waveform).

Metric	Value visible in report figure
Transmit count	4
Receive count	3
PASS count	3
FAIL count	0
Transmitted bytes shown	0x41, 0x55, 0xA5, 0xFF
Received bytes shown	0x41, 0x55, 0xA5

Capture limitation: The displayed figure shows only three received bytes. It does not show the fourth byte (0xFF) being received. Therefore, this figure supports 3 successful matches with 0 failures in the visible capture, not a completed 4/4 receive test.

2. Post-layout gate-level simulation

Source: internship report, page 33, Figure 9.1. The displayed post-layout waveform likewise shows TX count = 4, RX count = 3, PASS = 3 and FAIL = 0. The received bytes visible in the capture are 0x41, 0x55 and 0xA5; reception of 0xFF is not visible.

3. Static timing analysis (STA)

Source: internship report, page 32, Figure 8.1. The report figure shows two timing paths, both marked “slack (MET)”.

Timing path	Slack shown	Status
Minimum / hold path	3.36	slack (MET)
Maximum / setup path	7.90	slack (MET)

The supplied screenshot does not explicitly state a unit beside these slack values. WNS/TNS summary values are not visible in the provided figure.

4. Physical design evidence

Evidence	Report location
Fully routed layout	Page 30, Figure 7.7
Final GDSII layout in KLayout	Page 35, Figure 10.3

5. Items not quantified in the supplied report

The supplied report includes DRC and LVS discussion/captions on page 34, but does not show the actual DRC/LVS run outputs or pass/fail counts. The abstract and conclusion also retain placeholders for final utilisation, WNS/TNS and DRC/LVS status.

Do not treat captions or explanatory text as proof of DRC/LVS sign-off. Add the actual tool-generated reports and values when available.

Source note

This summary records only what is visible in the supplied internship report: pages 24, 30, 32, 33, 34 and 35. It is not a fresh simulator, synthesis, STA, routing, DRC or LVS run. Raw VCD files, simulator logs, synthesis reports, OpenROAD outputs and KLayout sign-off reports were not included in the source material.